

Polwatta Gallage, Inuka Jithmal

Electrical Engineering Undergraduate

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PROFILE

Innovative developer and engineering student specializing in AI agent architectures and full-stack applications. Proven ability to design, build, and deploy complex systems using Java, Spring Boot, and generative AI API integrations. A methodical problem-solver with a passion for building intelligent software solutions and bridging the gap between cutting-edge technology and practical, user-centric applications.

EDUCATION

- ▶ **University of Moratuwa** — B.Sc. (Hons) in Electrical Engineering
 - Current CGPA: 3.40 / 4.00 (up to 4/8 semesters)
 - Relevant Coursework: Data Structures and Algorithms, Modular Software Development
- ▶ **Rahula College Matara** — G.C.E. Advanced Level Examination, Physical Science Stream
 - Achieved AAA (Distinction Passes)
 - Island Rank 311, District Rank 26

SKILLS

- ▶ **Programming** — Python, C++, Java
- ▶ **Tools** — GitHub, Altium, SolidWorks, EasyEDA, MATLAB, LTspice, Overleaf (LaTeX)
- ▶ **Soft Skills** — Teamwork, Leadership, Interpersonal Communication, Public Speaking, Organization, Problem Solving

PROJECTS

Alga-Bot Smart Photobioreactor System (EE2043) — *Embedded Systems, PCB Design, Full-Stack*

- Designed a full-stack ecosystem integrating custom LM358 signal-conditioning PCBs with LabVIEW data acquisition
- Integrated a Python FastAPI backend, Raspberry Pi AI agents, and environmental sensors via HTTP communication
- Developed real-time web monitoring, automated PDF reporting, and AI decision-making for microalgae cultivation
- *Contribution:* System architecture, PCB design, full-stack software integration, AI agent implementation

 [GitHub](#)

Advanced Light Intensity Indicator (ALII) — *Analog Electronics, Digital Logic Design*

- Designed a custom ALII module using purely hardware-based, discrete gate-level logic for maximum efficiency
- Integrated Flash ADC inputs, adders, and flip-flops for an accumulator-based, shift-division circuit
- Developed a drift-free, digital block-averaging architecture for signal integrity over analog integrators
- *Contribution:* System architecture, digital logic design, hardware strategy, viva defense preparation

 [GitHub](#)

IoT-Based Smart Energy Monitoring System (SEMS) — *IoT, PCB Design, SolidWorks, MATLAB*

- Integrated MATLAB/Simulink and MQTT for real-time, appliance-level energy monitoring
- Designed a secure two-layer PCB with strict isolation to detect standby phantom loads
- Modeled an ABS enclosure in SolidWorks, validating 24/7 fanless operation at 28°C
- *Contribution:* Hardware architecture, PCB design, 3D modeling, thermal analysis

 [GitHub](#)

Autonomous Path-Finding Robot (PeraBots 25 Finalist) — *Robotics, Webots Simulation*

- Engineered a Webots LIDAR obstacle avoidance system using real-time sensor feedback
- Integrated path optimization algorithms to maximize navigational speed and accuracy
- Developed dynamic track-learning logic to eliminate redundant paths and reduce lap times
- *Contribution:* Simulation modeling, LIDAR integration, path optimization algorithms

 [GitHub](#)

G.W.E.N. 5.0 (Generative Workflow & Execution Nexus) — *AI, Full-Stack Development*

- Developed an end-to-end AI suite using React Native and Python for system logic and document compilation
- Engineered a dynamic PDF reporting engine embedding contextually relevant, AI-generated images
- Integrated advanced generative AI APIs to drive complex workflows and voice assistant capabilities
- *Contribution:* Full-stack development, AI API integration, automated reporting logic, frontend UI/UX design

 [GitHub](#)  [Presentation](#)

Rice Saver Grain Protection System (IEEE Arduino Top 10) — *Embedded Systems, Arduino, Automation*

- Designed an Arduino-based system to monitor storage temperature and humidity for spoilage prevention.
- Engineered a thermal pest control cycle and integrated PIR sensors for real-time intrusion detection.
- Implemented an I2C LCD dashboard to display real-time environmental diagnostics and system warnings.
- *Contribution:* Hardware architecture, circuit integration, sensor interfacing, and control logic.

 [GitHub](#)

Task-Performing Robot (SLRC 2025) — *Robotics, Embedded Systems, Hardware Development*

- Built a fully functional, task-performing robot for the SLRC 2025 at the University of Moratuwa
- Integrated complex systems, translating theoretical schematics into a responsive physical machine
- Conducted rigorous hardware troubleshooting and mechanical debugging under competitive conditions
- *Contribution:* Hardware development, system integration, real-time mechanical troubleshooting

 [GitHub](#)

IoT-Based Smart Bicycle Lock System — *IoT, Embedded Systems, Wireless Comm*

- Designed an IoT smart bicycle lock featuring secure, remote smartphone actuation
- Engineered an anti-theft mechanism that sends instant alerts and real-time GPS locations
- Integrated wireless modules and hardware sensors with a microcontroller for robust automation
- *Contribution:* Embedded architecture, sensor integration, anti-theft automation logic

 [GitHub](#)

ACHIEVEMENTS & AWARDS

- › Selected as a Finalist in RoboFest 2025 for the Micromouse Competition, organized by SLIIT
- › Selected as a Top 10 Finalist in the IEEE Arduino Challenge Sphere in 2024
- › Participated in SLRC in 2025
- › Participated in Perabots in 2025 and became a semifinalist

VOLUNTEERING

- › Served as a Co-chair for EESpire 2026, University of Moratuwa
- › Served as Program Lead of RoboRoarZ 2026, University of Moratuwa
- › Serving as Co-Chair for the “Atomic Guider” project, IEEE Nuclear and Plasma Sciences Society, University of Moratuwa
- › Member of the Finance and Industry Relations Committee for IEEE Open Week 2026, IEEE Student Branch, University of Moratuwa
- › Served as Selection Committee Member for MoraForesight 3.0, University of Moratuwa
- › Served as Logistics Committee Member for the MERCon 2024 international conference
- › Member of Moraspirit, University of Moratuwa
- › Member of Gavel Club, University of Moratuwa
- › Served as Compere for the “Enlumen 2025” get-together, of EESoc

REFERENCES

Prof. Buddhika Jayasekara

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